



## Mitoprep Answer Key 2

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**Q1) Explanation:**

**Hint:**  $H_{\max} = \frac{u^2}{2g}$

**Step 1:** Find the net retardation.

$$\begin{array}{c} g \downarrow \\ 0.1g \downarrow \\ a = -1.1g \end{array}$$

**Step 2:** Find the maximum height.

$$h = \frac{u^2}{2g}$$
$$h' = \frac{u^2}{2(1.1g)}$$

**Step 3:** Find the percentage decrease in height.

$$\frac{h-h'}{h} \times 100 = \left( \frac{\frac{u^2}{2g} - \frac{u^2}{2(1.1g)}}{\frac{u^2}{2g}} \right) \times 100 = \frac{1}{1.1} \times 100 = \frac{100}{1.1} = 9\%$$

Therefore, the percentage decrease in height is 9%.

Hence, option (2) is the correct answer.

**Q2) Explanation:**

**Hint:** Write the position vector of the car.

**Step:** Find the displacement of the car.

$$\vec{r} = 3\hat{i} - 4\hat{j} - 6\hat{i} + 8\hat{j}$$

$$\Rightarrow \vec{r} = -3\hat{i} + 4\hat{j}$$

$$\Rightarrow |\vec{r}| = \sqrt{3^2 + 4^2}$$

$$\Rightarrow |\vec{r}| = 5 \text{ km}$$

Hence, option (1) is the correct answer.

**Q3) Explanation:**

**Hint:** At the highest point of the trajectory, only gravity acts on the projectile.

**Step:** Find the net force acting on a projectile at the highest point of its trajectory.

At the highest point of the trajectory, the velocity  $v$  is purely horizontal. There is no vertical component of velocity, so no force due to acceleration in the vertical direction other than gravity.

The gravitational force acting on the projectile is given by;

$$\Rightarrow F_{\text{gravity}} = mg,$$

where  $m$  is the mass of the projectile and  $g$  is the acceleration due to gravity. There is no other force acting on the projectile because no additional forces, such as air resistance, are mentioned in the problem. The net force acting on the projectile at the highest point is simply  $mg$ .

Hence, option (2) is the correct answer.

**Q4) Explanation:**

**Hint:** Average time for the first revolution is the total time to cover four revolutions divided by four.

**Step 1:** Find the time to cover each revolution.

Time to cover the first revolution,  $t_1 = 12 \text{ s}$

Time to cover the second revolution,  $t_2 = 6 \text{ s}$ ; as the speed doubles.

Time to cover the third revolution,  $t_3 = 4 \text{ s}$ ; as the speed triples.

Time to cover the fourth revolution,  $t_4 = 3 \text{ s}$ ; as the speed quadruples.

**Step 2:** Find the average time to cover four revolutions.

Total time to cover four revolutions,  $t = t_1 + t_2 + t_3 + t_4 = 12 + 6 + 4 + 3 = 25 \text{ s}$

The average time to cover four revolutions is given by;

$$t_{\text{avg}} = \frac{t}{4} = \frac{25}{4} = 6.25 \text{ s}$$

Hence, option (3) is the correct answer.

Q5) Explanation:

$$\text{Hint: } \vec{a}_{\text{avg}} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t}$$

**Step:** Find the average acceleration of the ball.

Considering downward quantities as positive and upward quantities as negative.

$$\begin{aligned} \vec{a}_{\text{avg}} &= \frac{\vec{v}_f - \vec{v}_i}{\Delta t} \\ &= \frac{v_2 - (-v_1)}{t_2 - t_1} \\ &= \frac{v_2 + v_1}{t_2 - t_1} \end{aligned}$$

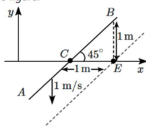
Hence, option (2) is the correct answer.

Q6) Explanation:

$$\text{Hint: Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

**Step:** Find the velocity of the point of intersection (C).

The displacement of the point of intersection after 1 s is 1 m along positive x-axis as shown in the figure.



Therefore the velocity of the intersection (C) of AB with x-axis, is 1 m/s along the positive x-axis.

Hence, option (1) is the correct answer.

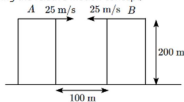
Q7) Not Possible - option (1) is correct answer

**Q8) Explanation:**

**Hint:** The bullets travel an equal distance in the horizontal direction.

**Step 1:** Find the relative velocity and acceleration.

Using the relative motion concept.



$$a_{A,B} = 0$$

$$u_{A,B} = 25 + 25 = 50 \text{ m/s along } x\text{-direction}$$

$$u_{A,B} = 0 \text{ along } y\text{-direction}$$

The meeting time is given by;

$$t = \frac{100}{50} = 2 \text{ s}$$

**Step 2:** Find the vertical displacements of the bullets.

In 2s, the vertical displacement of each bullet in the ground frame.

$$y = ut + \frac{1}{2}gt^2$$

Substituting the known values we get.

$$y = 0 + \frac{1}{2}gt^2$$

$$\Rightarrow y = \frac{1}{2} \times 10 \times 2^2$$

$$\Rightarrow y = 20 \text{ m}$$

Thus the height from the ground is given by;

$$\Rightarrow h = (200 - 20) = 180 \text{ m}$$

Hence, option **(1)** is the correct answer.

**Q9) Explanation:**

$$\text{Hint: } H = \frac{v^2 \sin^2 \theta}{2g}$$

**Step:** Find the relation between  $\theta_1$  and  $\theta_2$ .

It is given that  $H_1 > H_2$ .

$$\Rightarrow \frac{v^2 \sin^2 \theta_1}{2g} > \frac{v^2 \sin^2 \theta_2}{2g}$$

$$\Rightarrow \sin^2 \theta_1 > \sin^2 \theta_2$$

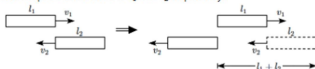
$$\Rightarrow \theta_1 > \theta_2$$

Hence, option **(1)** is the correct answer.

**Q10) Explanation:**

**Hint:** Use relative velocity.

**Step 1:** Find the relative velocity of the trains when the train crosses each other.  
Let the speed of the trains be  $v_1$  and  $v_2$  respectively.



$$v_{rel} = v_1 + v_2$$

$$S_{rel} = l_1 + l_2$$

As the trains cross each other in 10 s then,

$$10 = \frac{200}{v_1 + v_2}$$

$$\Rightarrow v_1 + v_2 = 20 \dots (1)$$

**Step 2:** Find the condition when trains overtake.



$$S_{rel} = l_1 + l_2$$

$$v_{rel} = v_2 - v_1$$

The time taken by the trains to overtake is 20 s i.e.,

$$\frac{S_{rel}}{v_{rel}} = \frac{l_1 + l_2}{v_2 - v_1} = 20$$

$$\frac{200}{v_2 - v_1} = 20$$

$$\Rightarrow v_2 - v_1 = 10 \dots (2)$$

**Step 3:** Find the speed of the trains.

From equations (1) and (2) we get;

$$v_2 - v_1 = 10$$

$$v_2 + v_1 = 20$$

After solving the above equations we get;

$$v_1 = 5 \text{ m/s}$$

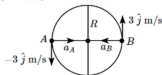
$$v_2 = 15 \text{ m/s}$$

Hence, option (3) is the correct answer.

**Q11) Explanation:**

**Hint:**  $a = \frac{v^2}{r}$

**Step 1:** Analyse the motion.



**Step 2:** Find the velocity at point  $A$ .

The velocity at point  $B$  is given,  $\vec{v}_B = 3\hat{j} \text{ m/s}$

The velocity of the particle at  $A$  is:

$$\vec{v}_A = -3\hat{j} \text{ m/s}$$

**Step 3:** Find the acceleration at point  $A$ .

The acceleration at this point  $B$  is given by;

$$\vec{a}_B = \frac{v_B^2}{R} = \frac{9}{1} = 9\hat{i} \text{ m/s}^2$$

The acceleration at the point  $A$  is  $\frac{v_A^2}{R}$

$$\vec{a}_A = 9\hat{i} \text{ m/s}^2.$$

Therefore, the velocity and acceleration at the point  $A$  is  $\vec{v}_A = -3\hat{j} \text{ m/s}$ ;  $\vec{a}_A = 9\hat{i} \text{ m/s}^2$

Hence, option (2) is the correct answer.

**Q12) Explanation:**

**Hint:**  $a = \frac{dv}{dt}$

**Step:** Find the correct variation of acceleration ( $a$ ) and time ( $t$ ).

The acceleration of the body is given by:

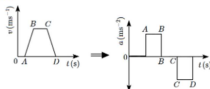
$$a = \frac{dv}{dt}$$

In the  $AB$  region, the velocity of the body is increasing uniformly means acceleration is positive and constant.

In the  $BC$  region, the velocity of the body is constant means acceleration is zero.

In the  $CD$  region, the velocity of the body is decreasing uniformly means acceleration is negative and constant.

Therefore, the velocity ( $v$ )-time ( $t$ ) and variation of acceleration ( $a$ ) and time ( $t$ ) are shown below.



Hence, option **(2)** is the correct answer.

**Q13) Explanation:**

**Hint:**  $\vec{v} = \vec{u} + \vec{a}t$

**Step:** Find the initial velocity of the car.

Given:  $\vec{a} = -0.1 \text{ m/s}^2$ ,  $t = 5 \text{ s}$ ,  $\vec{v} = 0 \text{ m/s}$

Let the initial velocity be  $\vec{u}$ .

According to the first equation of motion, the final velocity of the car is given by:  $\Rightarrow \vec{v} = \vec{u} + \vec{a}t$

$$\Rightarrow 0 = \vec{u} - 0.1 \times 5 \Rightarrow \vec{u} = 0.5 \text{ m/s}$$

Therefore, the initial velocity of the car is  $0.5 \text{ m/s}$ .

Hence, option **(3)** is the correct answer.

**Q14) Explanation:**

**Hint:** At point  $B$  particle will have only a horizontal velocity component.

**Step:** Find the velocity at the point  $B$ .

The horizontal component of the projectile remains constant, and at the highest point, the vertical component of the velocity is zero.

The velocity of the ball at the highest point is given by:

$$v_B = v \cos \theta$$

$$\Rightarrow v_B = 20 \cos 60^\circ$$

$$\Rightarrow v_B = 20 \times \frac{1}{2} = 10 \text{ ms}^{-1}$$

Hence, option **(4)** is the correct answer.

**Q15) Explanation:**

**Hint:**  $a = \frac{d^2\vec{r}}{dt^2}$

**Step:** Find the direction of acceleration of the particle.

The position vector of the particle is given as:

$$\vec{r}(t) = 8t\hat{i} + 5t^2\hat{j} + 6\hat{k}$$

The velocity vector of the particle is given by:

$$\vec{v} = \frac{d\vec{r}}{dt}$$
$$\Rightarrow \vec{v} = 8\hat{i} + 10t\hat{j}$$

The acceleration of the particle is given by:

$$\vec{a} = \frac{d\vec{v}}{dt}$$
$$\Rightarrow \vec{a} = 10\hat{j}$$

Therefore, the acceleration of the particle is along the positive  $y$ -axis.

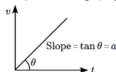
Hence, option **(1)** is the correct answer.

**Q16) Explanation:**

**Hint:** The graph is a straight line with a positive slope.

**Step:** Find the correct graph for the motion of the car.

The slope of the  $v$ - $t$  graph gives the acceleration as shown in the figure below.



(1) In the first graph, the slope is positive hence acceleration is positive.

(2) In the second graph, the velocity decreases with time hence acceleration is negative.

(3) In the third option, the velocity decreases with time hence acceleration is negative.

(4) In the fourth option, the velocity remains constant with time hence acceleration is zero.

Therefore, the graph for the uniform positive acceleration is shown in the figure below:



Hence, option **(1)** is the correct answer.

**Q17) Explanation:**

**Hint:** Distance covered = Length of the path

**Step 1:** Write the position equation.

$$x = 1 - 2t + t^2$$

**Step 2:** Find the velocity.

$$v = \frac{dx}{dt}$$
$$= 2t - 2$$

**Step 3:** Find a time when the particle changes direction.

$$v = 0$$

$$2t - 2 = 0$$

$$t = 1\text{ s}$$

**Step 4:** Find position at various times.

$$\text{At } t = 0 \text{ s, } x = 1 \text{ m}$$

$$\text{At } t = 1 \text{ s, } x = 0 \text{ m}$$

$$\text{At } t = 3 \text{ s, } x = 4 \text{ m}$$

$$\text{So the total distance covered by the particle} = 1 + 4 = 5 \text{ m}$$

Hence, option **(2)** is the correct answer.

**Q18) Explanation:**

**Hint:** The horizontal component of velocity in projectile motion remains constant.

**Step:** Find the speed of the projectile at the highest point.

The horizontal component of velocity at the highest point in projectile motion remains constant, and the vertical component of velocity at the highest point is zero.

Therefore, the speed of the projectile at the highest point is  $3 \text{ ms}^{-1}$ .

Hence, option (1) is the correct answer.

**Q19) Option (1) is correct**

**Q20) Explanation:**

**Hint:**  $|\vec{v}_1 - \vec{v}_2| = \sqrt{v_1^2 + v_2^2 - 2v_1v_2 \cos \theta}$

**Step:** Analyse each statement one by one.

(A) The magnitude of velocity is speed. Since the particle's speed is constant, the change in the magnitude of its velocity is zero. Hence, statement (A) is true.

(B) The change in the magnitude of its velocity is zero.

Hence, statement (B) is false.

(C) When a particle moves with constant speed  $v$  and changes its direction by an angle  $\theta$  as shown in the figure.



The speed remains unchanged, but the velocity vector changes direction.

The magnitude of change in velocity is given by:

$$\begin{aligned} |\vec{v}_1 - \vec{v}_2| &= \sqrt{v_1^2 + v_2^2 - 2v_1v_2 \cos \theta} \\ \Rightarrow |\Delta v| &= \sqrt{v^2 + v^2 - 2v^2 \cos \theta} \quad [\because v_1 = v_2 = v] \\ \Rightarrow |\Delta v| &= \sqrt{2v^2 - 2v^2 \cos \theta} \\ \Rightarrow |\Delta v| &= \sqrt{2v^2(1 - \cos \theta)} \\ \Rightarrow |\Delta v| &= \sqrt{4v^2 \sin^2 \left( \frac{\theta}{2} \right)} \quad \left[ \because 1 - \cos \theta = \sin^2 \left( \frac{\theta}{2} \right) \right] \\ \Rightarrow |\Delta v| &= 2v \sin \left( \frac{\theta}{2} \right) \end{aligned}$$

So, the magnitude of change in velocity is  $2v \sin \left( \frac{\theta}{2} \right)$ .

Hence, statement (C) is true.

(D) The magnitude of change in velocity is  $2v \sin \left( \frac{\theta}{2} \right)$ .

Hence, statement (D) is false.

Therefore, the correct statements are (A) and (C) only.

Hence, option (2) is the correct answer.

**Q21) Explanation:**

**Hint:**  $a = \omega^2 R$

**Step:** Find the order of the average acceleration of the particle.

The length of the minute hand is  $0.1 \text{ m}$ .

The angular frequency of the minute hand is given by;  $\omega = \frac{2\pi}{T} = \frac{2\pi}{60} = 0.105 \text{ rad/s}$

The average acceleration of the tip is given by;

$$a = \omega^2 R = (0.105)^2 (0.1) = 0.0011 = 1.1 \times 10^{-3}$$

Therefore, the average acceleration is of the order of  $10^{-3}$ .

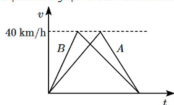
Hence, option (1) is the correct answer.



**Q22) Explanation:**

**Hint:** Plot the speed-time graph of the cars and compare them.

**Step 1:** Plot and analyze the speed vs time graphs for both the cars  
The speed-time graph for both cars is shown in the figure below;



**Step 2:** Find the relation between the time  $t_A$  and  $t_B$ .

The area of the triangle under the graph gives the total distance travelled. So, the distance travelled are same for both cars.

The total distance covered by the car is given by;

Total Distance =  $\frac{1}{2} \times \text{Base}(\text{total time}) \times \text{Height}(\text{max velocity})$

$$\Rightarrow \frac{1}{2} \times t_A \times 40 = \frac{1}{2} \times t_B \times 40$$

$$\Rightarrow t_A = t_B$$

Therefore, the time ( $t_A$ ) taken by the car  $A$  is equal to the time ( $t_B$ ) taken by the car  $B$ .

Hence, option (4) is the correct answer.

**Q23) Explanation:**

**Hint:** Average velocity =  $\frac{\text{Displacement}}{\text{time}}$

**Step:** Find the average velocity vector.

Given:

Position of particle at  $t = 0$  is  $2\hat{i} + 3\hat{j}$ .

At  $t = 2$  s is  $6\hat{i} + 7\hat{j}$ .

At  $t = 5$  s is  $13\hat{i} + 14\hat{j}$ .

Displacement between  $t_0$  and  $t_5$ , =  $11\hat{i} + 11\hat{j}$

$$\Rightarrow \text{Average velocity} = \text{Displacement} / \text{time} = \frac{(13\hat{i} + 14\hat{j}) - (2\hat{i} + 3\hat{j})}{5} = \frac{11(\hat{i} + \hat{j})}{5}$$

Hence, option (4) is the correct answer.

**Q24) Explanation:**

**Hint:** Velocity is defined as the rate of change of position.

**Step 1:** Find the velocity of car  $P$ .

The displacement of the car  $P$  is given as:  $x_P(t) = at + bt^2$

The velocity of the car  $P$  is given by;

$$v = \frac{dx_P}{dt}$$

$$\Rightarrow v_P = a + 2bt \dots (1)$$

**Step 2:** Find the velocity of car  $Q$ .

The displacement of the car  $Q$  is given as;

$$x_Q(t) = ft - t^2$$

The velocity of the car  $Q$  is given by;

$$\Rightarrow v_Q = f - 2t \dots (2)$$

**Step 3:** Find the time at which  $v_P = v_Q$ .

From equation (1) and (2) we get;

$$a + 2bt = f - 2t$$

$$\Rightarrow 2bt + 2t = f - a$$

$$\Rightarrow t = \frac{f-a}{2(b+1)}$$

Hence, option (4) is the correct answer.

**Q25) Explanation:**

**Hint:**  $\vec{v}_{MR} = \vec{v}_{BG} + \vec{v}_{MG}$

**Step:** Find the speed of (effort of) the man w.r.t., the river water.

$$\vec{v}_{MR} = \vec{v}_{BG} + \vec{v}_{MG}$$

$$\Rightarrow \vec{v}_{MR} > \vec{v}_{BG}$$

$$\Rightarrow \vec{v}_{MR} > 5 \text{ m/s}$$

Therefore, the speed of (effort of) the man w.r.t., the river water is greater than 5 m/s.

Hence, option **(3)** is the correct answer.

**Q26) Explanation:**

**Hint:**  $S = ut + \frac{1}{2}at^2$

**Step:** Find the appropriate position-time ( $x-t$ ) graph for positive accelerated motion.

From the second equation of kinematics, we have:

$$S = ut + \frac{1}{2}at^2$$

The position-time ( $x-t$ ) graph for positive acceleration is a curve that opens upwards. This means that the graph is concave up, indicating that the position of the particle increases at an increasing rate over time. Initially, the slope (which represents velocity) may be small but becomes steeper as time progresses, reflecting the increase in velocity due to positive acceleration.

The position-time ( $x-t$ ) graph for positive accelerated motion is shown in the figure below:

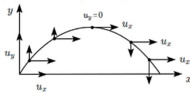


Hence, option **(1)** is the correct answer.

**Q27) Explanation:**

**Hint:** Speed is minimum at the highest point.

**Explanation:** When an object is launched from the ground with a certain velocity at an angle to the horizontal, its velocity gradually decreases until it reaches its maximum height, where it becomes minimal. Subsequently, as it descends, its velocity increases until it reaches the ground. This velocity profile of the object is depicted in the figure provided below.



Therefore, the particle's speed will be minimal at the highest point.

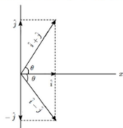
Hence, option **(3)** is the correct answer.

**Q28) Explanation:**

**Hint:** The magnitude of the unit vector is 1.

**Step:** Find the magnitude and direction of the vector  $(\hat{i} - \hat{j})$ .

The position vectors  $(\hat{i} + \hat{j})$  and  $(\hat{i} - \hat{j})$  are shown in the figure below;



The phase angle for the vector  $(\hat{i} + \hat{j})$  is given by

$$\tan \theta = \frac{y}{x} = \frac{1}{1} = 1$$

$$\Rightarrow \theta = 45^\circ$$

The magnitude of the vector  $(\hat{i} - \hat{j})$  is given by;

$$|\hat{i} - \hat{j}| = \sqrt{(1)^2 + (2)^2} = 2 \times 1 \times 1 \times \cos 90^\circ$$

$$\Rightarrow |\hat{i} - \hat{j}| = \sqrt{2} = 1.414 \text{ units}$$

The phase angle for the  $(\hat{i} - \hat{j})$  is given by;

$$\tan \theta = \frac{-y}{x} = \frac{-1}{1} = -1$$

$$\Rightarrow \theta = -45^\circ$$

Therefore, the vector  $(\hat{i} - \hat{j})$  makes an angle  $-45^\circ$  with the  $x$ -axis.

Hence, option **(2)** is the correct answer.

**Q29) Explanation:**

$$\text{Hint: } a_c = \frac{v^2}{r}$$

The centripetal acceleration of a particle moving on a circular path is given by:

$$a_c = \frac{v^2}{R}$$

In the given figure,

$$a_c = a \cos 30^\circ = 15 \cos 30^\circ \text{ m/s}^2$$

$$\frac{v^2}{R} = 15 \cos 30^\circ$$

$$\therefore v = 5.7 \text{ m/s}$$

**Q30) Explanation:**

**Hint:** Acceleration is defined as the change of velocity divided by the time interval.

**Step:** Analyse each option one by one.

1. Any motion in a straight line: There are two motions possible in this case. One with constant velocity motion which can have zero acceleration however the other motion is of changing speed (non-uniform motion), which involves acceleration. Thus, not all straight-line motions have zero acceleration.

2. Projectile motion: This always involves acceleration due to gravity acting on the projectile. Even if the horizontal component of the velocity is constant, the vertical component changes due to gravity, resulting in a non-zero acceleration.

3. Any motion in a circle: Circular motion involves centripetal acceleration, which is always directed toward the center of the circle. Hence, any object moving in a circular path has a non-zero acceleration.

Therefore, none of the motions above guarantees zero acceleration.

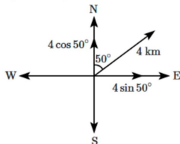
Hence, option **(4)** is the correct answer.

**Q31) Explanation:**

**Hint:** Resolve the components of the elephant's displacement.

**Step:** Find the components of the elephant's displacement.

The elephant's displacement is shown in the figure below;



The displacement of the elephant in the easterly direction is  $4 \sin 50^\circ = 3.06$  km

The displacement of the elephant in the northerly direction is  $4 \cos 50^\circ = 2.57$  km

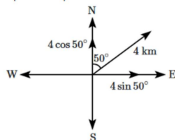
Hence, option (1) is the correct answer.

**Q31) Explanation:**

**Hint:** Resolve the components of the elephant's displacement.

**Step:** Find the components of the elephant's displacement.

The elephant's displacement is shown in the figure below;



The displacement of the elephant in the easterly direction is  $4 \sin 50^\circ = 3.06$  km

The displacement of the elephant in the northerly direction is  $4 \cos 50^\circ = 2.57$  km

Hence, option (1) is the correct answer.

**Q32) Explanation:**

**Hint:** Relate  $x$  and  $y$  by substituting  $t$ .

**Step:** Find the initial velocity of the projectile at  $t = 0$ .

Given:  $x = 36t$ ,  $y = 48t - 4.9t^2$

The initial velocity of the particle at  $t = 0$  is given by;

$$u_x = \frac{dx}{dt} = 36 \text{ m/s}$$

$$u_y(t=0) = \frac{dy}{dt} = 48 - 9.8t = 48 \text{ m/s}$$

**Step 2:** Find the angle of the projection.

The angle of the projectile is given by;

$$\tan \theta = \frac{u_y}{u_x} = \frac{48}{36} = \frac{4}{3}$$

$$\Rightarrow \theta = \tan^{-1} \left( \frac{4}{3} \right)$$

$$\Rightarrow \sin \theta = \frac{4}{5}$$

$$\Rightarrow \theta = \sin^{-1} \left( \frac{4}{5} \right)$$

Hence, option (1) is the correct answer.

**Q33) Explanation:**

**Hint:** The average velocity of an object is the ratio of the displacement and the corresponding time interval.

**Step:** Find the average velocity of the car.

The average velocity of the car is given by:

$$\begin{aligned} \frac{n}{v_{\text{avg}}} &= \frac{1}{v_1} + \frac{1}{v_2} + \dots + \frac{1}{v_n} \\ \Rightarrow \frac{3}{v_{\text{avg}}} &= \frac{1}{11} + \frac{1}{2 \times 11} + \frac{1}{3 \times 11} \\ \Rightarrow \frac{3}{v_{\text{avg}}} &= \frac{6+3+2}{66} = \frac{11}{66} \\ \Rightarrow \frac{3}{v_{\text{avg}}} &= \frac{1}{6} \end{aligned}$$

$$\Rightarrow v_{\text{avg}} = 18 \text{ m/s}$$

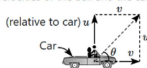
Hence, option **(2)** is the correct answer.

**Q34) Explanation:**

**Hint:**  $\tan \theta = \frac{v_y}{v_x}$

**Step:** Find the correct relation between the  $u$ ,  $v$  and  $\theta$ .

The velocities of the ball and the car are shown in the figure below:



From the above diagram:

$$\begin{aligned} \tan \theta &= \frac{u}{v} \\ \Rightarrow \frac{\sin \theta}{\cos \theta} &= \frac{u}{v} \\ \Rightarrow v \sin \theta &= u \cos \theta \end{aligned}$$

Hence, option **(2)** is the correct answer.

**Q35) Explanation:**

**Hint:** Distance  $\geq$  |Displacement|

**Step:** Find the ratio of average velocity and average speed.

$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time}}$$

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$\frac{\text{Average velocity}}{\text{Average speed}} = \frac{\text{Displacement}}{\text{Distance}} \leq 1$$

Therefore, the ratio may be equal or less than 1.

Hence, option **(4)** is the correct answer.

**Q36) Explanation:**

**Hint:**  $\vec{v}_{R,M} = \vec{v}_R - \vec{v}_M$

**Explanation:** If the umbrella is held vertically and the man is still getting wet, it implies that the rain is coming from behind him, meaning the rain is not vertical but rather is being pushed towards him due to man forward motion.

Therefore, if the umbrella is vertical and the man is running forward, then the rain must be coming from the back.

Hence, option **(3)** is the correct answer.

**Q37) Explanation:**

**Hint:**  $a_c = \frac{v^2}{r}$

**Step 1:** Find the velocity at  $t = 2$  s.

$$\vec{v} = (2t\hat{i} + 3t^2\hat{j}) \text{ m/s}$$

At,  $t = 2$  s

$$\vec{v} = (4\hat{i} + 12\hat{j}) \text{ m/s}$$

$$\Rightarrow |\vec{v}| = \sqrt{16 + 144} = \sqrt{160} \text{ m/s}$$

**Step 2:** Find the centripetal acceleration of the particle.

The centripetal acceleration of the particle is given by:  $a_c = \frac{v^2}{r}$

$$\Rightarrow a_c = \frac{v^2}{r} = \frac{160}{2} \text{ m/s}^2$$

$$\Rightarrow a_c = 80 \text{ m/s}^2$$

Therefore, the centripetal acceleration of the particle is  $80 \text{ m/s}^2$ .

Hence, option (1) is the correct answer.

**Q38) Explanation:**

**Hint:** The slope of x-t graph represents the velocity.

**Step:** Find the velocity of the particle.

Initially, the velocity of the particle is constant and positive and it changes abruptly to negative during the motion.

**Q39) Explanation:**

**Hint:**  $R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$

**Step 1:** Find the radius of the circular path.

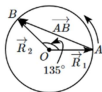
The distance travelled by the body on the circular path is given by:

$$\Rightarrow \frac{\theta}{360^\circ} \times 2\pi R = 60$$

$$\Rightarrow \frac{135^\circ}{360^\circ} \times 2\pi R = 60$$

$$\Rightarrow R = \frac{80}{\pi}$$

**Step 2:** Find the magnitude of the displacement of the person.



$$\vec{R}_2 = \vec{R}_1 + \vec{AB}$$

$$\Rightarrow \vec{AB} = \vec{R}_2 - \vec{R}_1$$

$$\Rightarrow |\vec{AB}| = |\vec{R}_2 - \vec{R}_1|$$

$$\Rightarrow |\vec{AB}| = \sqrt{R^2 + R^2 - 2R^2 \times \cos 135^\circ}$$

$$\Rightarrow |\vec{AB}| = \sqrt{R^2 + R^2 - 2R^2 \times (-0.7)}$$

$$\Rightarrow |\vec{AB}| = \sqrt{2R^2 + 1.4R^2}$$

$$\Rightarrow |\vec{AB}| = \sqrt{3.4R}$$

$$\Rightarrow |\vec{AB}| = 1.84 \times \frac{80}{3.14}$$

$$\Rightarrow |\vec{AB}| = 46.9 \approx 47 \text{ m}$$

Hence, option (2) is the correct answer.

**Q40) Explanation:**

**Hint:**  $v_0 = \sqrt{2gh}$

**Step:** Find the initial speed of the ball.

By using the third equation of motion we get;

$$v^2 - u^2 = 2as$$

$$0 - v_0^2 = 2(-9.8) \times 10$$

$$\Rightarrow v_0 = \sqrt{196}$$

$$= 14 \text{ ms}^{-1}$$

Therefore, the initial speed of the ball is  $14 \text{ ms}^{-1}$ .

Hence, option (3) is the correct answer.

**Q41) Explanation:**

**Hint:**  $v^2 = u^2 + 2as$

**Step 1:** Find the velocity just before hitting ground.

$$v^2 = u^2 + 2gh$$

$$v^2 = 2g(100) \dots(i)$$

**Step 2:** Find the displacement when velocity is  $\frac{v}{2}$ .

$$\left(\frac{v}{2}\right)^2 = 2gH$$

From equations (i) and (ii), we get:

$$\frac{1}{4} = \frac{H}{100}$$

$$H = 25 \text{ m}$$

**Step 3:** Find the height from the ground.



So, the height from the ground =  $(100 - 25) = 75 \text{ m}$ .

**Q42) Explanation:**

**Hint:**  $s = ut + \frac{1}{2}at^2$

**Step 1:** Find the time taken to reach the bottom.



$$s = \frac{1}{2}at^2$$

$$t = \sqrt{\frac{2s}{a}} = 4 \dots\dots(i)$$

**Step 2:** Find the time taken to travel one-fourth distance.

$$\frac{s}{4} = \frac{1}{2}a(t')^2$$

$$t' = \left[ \sqrt{\frac{2s}{a}} \right] \frac{1}{2} = \frac{t}{2} = 2 \text{ s}$$

Hence, option (2) is the correct answer.

**Q43) Explanation:**

**Hint:** The x-component of velocity remains constant throughout the projectile motion.

**Step:** Find the speed of the particle at a given position.

The horizontal component of the velocity of the projectile remains constant.

$$v \cos \theta = v_0 \cos \theta_0$$

$$\Rightarrow v = v_0 \cos \theta_0 \sec \theta$$

$$\Rightarrow v = A \sec \theta$$

Where  $A$  is the constant and equal to the initial horizontal component of the velocity.

Hence, option (4) is the correct answer.

**Q44) Explanation:**

$$\text{Hint: } s = ut + \frac{1}{2}at^2$$

**Step:** Find the time taken to reach the height of 15 m above the ground.

$$s = ut + \frac{1}{2}at^2$$

$$15 = 20t - \frac{1}{2} \times 10 \times t^2$$

$$t^2 - 4t + 3 = 0$$

$$\Rightarrow t = 1 \text{ s or } 3 \text{ s}$$

Hence, option (3) is the correct answer.

**Q45) Explanation:**

$$\text{Hint: } R = \frac{v^2 \sin 2\theta}{g}$$

**Step 1:** Find the range of the shell.

$$R = \frac{v^2 \sin 2\theta}{g}$$

$$\sin 2\theta = \frac{gR}{v^2}$$

$$\theta = \frac{1}{2} \sin^{-1} \left[ \frac{gR}{v^2} \right]$$

**Q46) Explanation:**

**Hint:** According to 1st law of thermodynamics,  $\Delta U = q + W$

**Explanation:**

Step 1: The given process is isothermal, i.e.,  $\Delta T = 0$

So, the internal energy change is  $\Delta U = C_V \Delta T$ . Thus, the internal energy change will be:  $\Delta U = 0$

Step 2: Also, vacuum conditions are given. Hence,  $p_{ext} = 0$

The work done will be  $(W) = -p_{ext} \Delta V = 0$

Thus, the heat will be:  $0 = q + 0$ ;  $q = 0$

Therefore, the work done and heat will also be zero. So, **option 3** is the correct answer.

- The gas is expanding **into a vacuum**, meaning there is no external pressure opposing the expansion.
- This is known as **free expansion** (or **Joule expansion**), where the gas expands without doing any work on the surroundings.



Q47) Step 1:

The given reaction is as follows:



For the given reaction  $\Delta n_g$  value is negative. Thus,  $\Delta S$  value is negative.

The bond formation is an exothermic reaction. So,  $\Delta H$  value is negative.

Step 2:

Bond formation = exothermic process

$$\Delta H = -Ve$$

$$\Delta S = -Ve$$

Hence, option third is the correct answer.

Q48) Explanation:

$$\text{Hint: } \Delta_{\text{Sol}} H^\ominus = \Delta_{\text{Lattice}} H^\ominus + \Delta_{\text{Hyd}} H^\ominus$$

Given:

- Lattice enthalpy ( $\Delta H_{\text{lattice}}$ ) of NaCl = +788 kJ/mol
- Enthalpy of solution ( $\Delta H_{\text{solution}}$ ) of NaCl = +4 kJ/mol

We need to find the hydration enthalpy ( $\Delta H_{\text{hydration}}$ ).

Rearranging the equation to solve for  $\Delta H_{\text{hydration}}$ :

$$\Delta H_{\text{hydration}} = \Delta H_{\text{solution}} - \Delta H_{\text{lattice}}$$

Substitute the given values:

$$\Delta H_{\text{hydration}} = 4 \text{ kJ/mol} - 788 \text{ kJ/mol}$$

$$\Delta H_{\text{hydration}} = -784 \text{ kJ/mol}$$

Q49) Explanation:

Hint:  $q_p = \Delta H$

Given that, 1 mol of  $\text{CCl}_4 = 154 \text{ g}$

$$\Delta_{\text{vap}} H \text{ for } 154 \text{ g } \text{CCl}_4 = 30.5 \text{ kJ}$$

$$\therefore \Delta_{\text{vap}} H \text{ for } 284 \text{ g } \text{CCl}_4 = \frac{30.5 \times 284}{154} \text{ kJ} = 56.25 \text{ kJ}$$

Q50) Explanation:

**Hint:** During an adiabatic process, the container should be insulator.

**Explanation:**

In an adiabatic process, no heat energy is exchanged between the gas and its surroundings. The gas does work on the environment during an expansion process, so its internal energy decreases. This means that the temperature of the gas will also decrease.

An adiabatic process is a thermodynamic process where a system is insulated from its surroundings, preventing heat from entering or leaving. A perfect conductor is an idealized material with infinite electrical conductivity and zero resistivity.

Hence, both assertion and reason are false. Thus, **option 4** is the correct answer.

Q51) Explanation:

$$\begin{aligned}\Delta H^{\circ} &= \frac{3}{2} \times \Delta H_1^{\circ} + \frac{\Delta H_2^{\circ}}{2} \\ &= -713.7 - 177.5 \\ &= -891.2 \text{ kJ}\end{aligned}$$

Q52) Explanation:

Hint: Follow the third law of thermodynamic

The third law of thermodynamics states that the entropy of a system approaches a constant value as the temperature approaches absolute zero.

The entropy of any pure crystalline substance approaches zero as the temperature approaches absolute zero.

Hence, option 4th is the correct answer.

Q53) Explanation:

Hint: The formula of work done is  $W = -nRT \ln \frac{V_f}{V_i}$

Step 1:

Given that, the work of reversible expansion under isothermal conditions can be calculated by using the expression

$$W = -nRT \ln \frac{V_f}{V_i}$$

$$V_f = 10 \quad V_i$$

$$T_2 = 600 \quad \text{K}$$

$$T_1 = 300 \quad \text{K}$$

Step 2:

Putting these values in the above expression

$$W_{600\text{K}} = -1 \times R \times 600 \quad \text{K} \quad \ln \frac{10}{1}$$

$$W_{300\text{K}} = -1 \times R \times 300 \quad \text{K} \quad \ln \frac{10}{1}$$

$$\text{Ratio} = \frac{W_{600\text{K}}}{W_{300\text{K}}} = \frac{1 \times R \times 600 \quad \text{K} \quad \ln \frac{10}{1}}{1 \times R \times 300 \quad \text{K} \quad \ln \frac{10}{1}} = \frac{600}{300} = 2$$

For the isothermal expansion of ideal gases,  $\Delta U = 0$ . Since temperature is constant this means there is no change in internal energy.

Q54) Explanation:

Given:

$$\Delta_f H^{\circ} \text{P}_4\text{O}_{10}(s) = -3110 \text{ kJ/mol}$$

$$\Delta_f H^{\circ} \text{H}_2\text{O}(l) = -286 \text{ kJ/mol}$$

$$\Delta_f H^{\circ} \text{H}_3\text{PO}_4(s) = -1279 \text{ kJ/mol}$$

Use formula:

$$\Delta_r H^{\circ} = \sum \Delta_f H^{\circ}(\text{products}) - \sum \Delta_f H^{\circ}(\text{reactants}) \dots (1)$$

**Step 2:** Calculate  $\sum \Delta_f H^{\circ}$  for products and reactants

For Products:

$$\sum \Delta_f H^{\circ}(\text{products}) = 4 \times (-1279) = -5116 \text{ kJ}$$

For Reactants:

$$\sum \Delta_f H^{\circ}(\text{reactants}) = (-3110) + 6 \times (-286) = -3110 - 1716 = -4826 \text{ kJ}$$

**Step 3:** put values in formula (1) to calculate  $\Delta_r H^{\circ}$

$$\Delta_r H^{\circ} = \sum \Delta_f H^{\circ}(\text{products}) - \sum \Delta_f H^{\circ}(\text{reactants})$$

$$\Delta_r H^{\circ} = -5116 - (-4826) = -5116 + 4826 = -290 \text{ kJ}$$

Hence, the **first option** is the correct option.

Q55) **Hint:** For a spontaneous process,  $\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{sur}} > 0$

**Explanation:**

The process of liquid water  $H_2O(l)$  turning into solid ice  $H_2O(s)$  is spontaneous at low temperatures. This is because the entropy decrease from the formation of a more ordered solid phase outweighs the enthalpy change.

Liquid is converting into solids; in the process, heat is released, therefore  $\Delta S_{\text{sys}} < 0$  and  $\Delta S_{\text{total}} > 0$

So, **option 1** is the correct answer.

Q56) **Explanation:**

**Hint:**  $\Delta G$  and  $\Delta S$  can be used to discriminate between reversible and irreversible process.

**Explanation:**

(i) When used in a reaction, Delta G is the difference in free energy between the products and the reactants. If this value is negative, the reaction is spontaneous, if it is positive, the reaction is non-spontaneous, and if it is zero, the reaction is at equilibrium.

(ii) The second law of thermodynamics states that any spontaneously occurring process will always lead to an escalation in the entropy ( $S$ ) of the universe. In simple words, the law explains that an isolated system's entropy will never decrease over time. It also explains, in exothermic reactions heat released by the reaction increases the disorder of the surroundings and overall entropy change is positive which makes the reaction spontaneous.

(iii)  $\Delta G$  and  $\Delta S$  can be used to discriminate between the reversible and irreversible process. This defines the spontaneity of the process.

For example, both for reversible and irreversible expansion for an ideal gas, under isothermal conditions,  $\Delta U = 0$ , but  $\Delta S_{\text{total}}$  i.e.,  $\Delta S_{\text{sys}} + \Delta S_{\text{sur}}$  is not zero for an irreversible process.

Thus,  $\Delta U$  does not discriminate between reversible and irreversible process, whereas  $\Delta S$  does.

(iv) The enthalpy of formation is the standard reaction enthalpy for the formation of the compound from its elements (atoms or molecules) in their most stable reference states at the chosen temperature (298.15K) and at 1bar pressure.

So, the heat released in the reaction  $CaO(s) + CO_2(g) \rightarrow CaCO_3(s)$  is not the enthalpy of the formation of  $CaCO_3$

Thus, option 3 is the correct answer.

Q57) **Explanation:**

**Hint:** Area under the curve represents work done.

In the PV graph, the area under the curve represents work done. In the fourth graph, volume is constant. Hence, work done is zero.

The graph represent in option third has the maximum work done because it covered the maximum area.

Thus, option third is the answer.



**Q58) Explanation:**

**Hint:** Use Hess's Law

**Explanation:**

Step 1: The given reaction is as follows:  $\text{F}_2\text{O}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightarrow \text{O}_2(\text{g}) + 2\text{HF}(\text{g})$

The heat of formation of  $\text{F}_2\text{O}(\text{g})$ ,  $\text{H}_2\text{O}(\text{g})$  and  $\text{HF}(\text{g})$  are 5.5 kcal, -57 kcal, and -64 kcal respectively.

This question aims to calculate the heat of the following reaction.

The heat of formation of  $\text{O}_2(\text{g})$  is 0 kcal. The standard enthalpy of formation of an element in its standard state is zero.

Step 2: The heat of the reaction is:  $\sum H_{\text{products}} - \sum H_{\text{reactants}}$

So, the heat of reaction =  $2 H_f(\text{HF}) - [H_f(\text{F}_2\text{O}) + H_f(\text{H}_2\text{O})]$

The heat of the reaction is:  $2(-64 \text{ kcal}) - [(5.5 \text{ kcal}) + (-57 \text{ kcal})]$  or

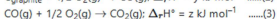
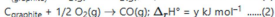
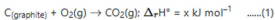
The heat of the reaction =  $(-128 + 51.5) \text{ kcal} = -76.5 \text{ kcal}$

So, the correct answer is **option 1**.

**Q59) Explanation:**

**Hint:** Follow Hess's Law

The given reactions are as follows:



If we add equation 2 and 3 we get equation 1. The same is done with the enthalpy also.

$$x = y + z$$

Thus, option second is the correct answer.

Q60)

Hint: The formula is  $\Delta S_{\text{fusion}} = \frac{\Delta H_{\text{fusion}}}{T}$

Step 1:

The given values are as follows:

$$\Delta H_{\text{fusion}} = 80 \text{ cal g}^{-1}$$

Temperature = 273 K

$$\Delta S_{\text{fusion}} = 28.80 \text{ cal K}^{-1}$$

Step 2:

Calculate the number of moles ( $x$ ) of ice melted as follows:

$$\begin{aligned}\Delta S_{\text{fusion}} &= \frac{\Delta H_{\text{fusion}}}{T} = \frac{80 \text{ cal g}^{-1} \times 18 \text{ g mol}^{-1}}{273 \text{ K}} \\ 28.80 \text{ cal K}^{-1} &= \frac{80 \text{ cal} \times 18 \text{ g}}{273} \text{ K}^{-1} \text{ mol}^{-1} \\ x &= \frac{273 \times 28.80}{80 \times 18} \text{ mol} \\ x &= 5.46 \text{ mol}\end{aligned}$$

Q61)

**Hint:** The volume of an ideal gas molecule is zero with respect to the gas.

**Explanation:**

The heat absorbed during the isothermal expansion of an ideal gas against a vacuum is zero.

During isothermal change, the temperature remains constant. Hence  $dT = 0$

We know that,  $dU = C_V dT$

Thus, at isothermal conditions,  $dU$  is zero.

One of the key assumptions of the ideal gas model is that the molecules of the gas have zero volume. However, for a non-ideal gas, the volume occupied is non-zero.

Thus, both the assertion and reason are correct, but reason is not the correct explanation for the assertion.

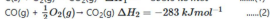
Hence, **option 2** is the correct answer.

**Q62) Explanation:**

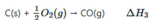
**Hint:** Use Hess's Law of Constant Heat Summation

Step 1:

The combustion of carbon and carbon monoxide is as follows:



The formation reaction of CO is as follows:



Step 2:

Reverse equation second and add with equation 1, the same scenario is done with enthalpy also.

$$\Delta H_3 = \Delta H_2 + \Delta H_1$$

$$\Delta H_3 = 283 + (-393.5)$$

$$= -110.5 \text{ kJ mol}^{-1}$$

**Q63) Explanation:**

**Hint:** Liquid is converted into solid, and system entropy decreases.

**Explanation:**

(i) Entropy is the degree of disorder or uncertainty in the system. The order of randomness is:

Gas > Liquid > Solid

(ii) When water freezes in a glass beaker,  $\Delta S_{\text{system}}$  decreases because molecules in solid ice are less random than in liquid water.

However, when water freezes to ice, heat is released, which is absorbed by the surroundings. Hence, the entropy of the surroundings ( $\Delta S_{\text{surroundings}}$ ) increases.

So, **option 3** is the correct answer.

**Q64) Hint:** A process or change is said to be reversible, if a change is brought out in such a way that the process could, at any moment, be reversed by an infinitesimal change.

**Explanation:**

A reversible process is defined as a process in which the system and surroundings can be returned to the original conditions from the final state without producing any changes in the thermodynamic properties of the universe if the process is reversed.

If a change is brought out in such a way that the process could, at any moment, be reversed by an infinitesimal (extremely small) change.

A reversible process proceeds infinitely slowly by a series of equilibrium states such that the system and the surroundings are always in near equilibrium with each other.

Hence, both statements are correct. Thus, **option 1** is the correct option.

**Q65) Explanation:**

Hint: Use the formula, that is,  $W = -2.303nRT \log \frac{P_1}{P_2}$

The given diagram represent that the process is carried out in infinite steps, hence it is isothermal reversible expansion of the ideal gas from pressure 2.0 atm to 1.0 atm 298 K.

$$W = -2.303 \quad nRT \quad \log \frac{P_1}{P_2}$$

$$W = -2.303 \times 1 \quad \text{mol} \times 8.314 \quad \text{JK}^{-1} \text{mol}^{-1} \times 298 \quad \text{K} \quad \log 2 \quad \left( \because \frac{P_1}{P_2} = \frac{2}{1} \right)$$

$$W = -2.303 \times 1 \times 8.314 \times 298 \times 0.3010 \quad \text{J}$$

$$W = -1717.46 \quad \text{J}$$

**Q66) Explanation:**

Hint: An isolated system cannot exchange either matter or energy with its surroundings.

**2. Isolated system**

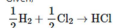
An isolated system cannot exchange either matter or energy with its surroundings.

**Q67)**

**Q68) Explanation:**

Hint:  $\Delta H_f = B \cdot E_{(\text{reactants})} - B \cdot E_{(\text{products})}$

Given,



$$\Delta H_f(\text{HCl}) = -93 \text{ kJ/mol}$$

Let the bond dissociation energy of H-Cl bond is x. For given reaction

$$\Delta H_f = \sum B \cdot E_{(\text{reactants})} - \sum B \cdot E_{(\text{products})}$$

$$-93 = \frac{1}{2}(434) + \frac{1}{2}(242) - x$$

$$x = 431 \text{ kJ/mol}$$

Hence, the correct option is the third option.

**Q69) Explanation:**

Hint:  $W = -P_{\text{ext}} (\Delta V)$

$$W = -P_{\text{ext}} (\Delta V)$$

$$= -8 \times 10^5 \text{ N/m}^2 \times (10 \times 10^{-3} \text{ m}^3)$$

$$= -8 \times 10^5 \times 10^{-2} \text{ joule}$$

$$= -8 \times 10^3$$

$$= -8000 \text{ J}$$

$$q + W = \Delta E$$

$$q + W = 0 \Rightarrow q = -W = +8000 \text{ J}$$

**Q70) Explanation:**

A spontaneous process is one where the randomness of the molecules in the system increases. Entropy is a scientific concept that describes how energy is distributed within a system and is often associated with disorder, randomness, or uncertainty.

The second law of thermodynamics states that the entropy of an isolated system will not decrease over time and will instead evolve towards thermodynamic equilibrium.

The Gibbs free energy change ( $\Delta G$ ) also determines spontaneity and is related to total entropy:

$$\Delta G = -T \Delta S_{\text{total}}$$

For spontaneous process,  $\Delta S_{\text{total}} > 0$ . Hence, **option 4** is the correct answer.

**Q71) Explanation:**

Step I:

Calculate the value of  $\Delta H$  as follows:

$$\begin{aligned}\Delta H &= \Delta U + \Delta n_g RT \\ &= 2100 + 2 \times 2 \times 300 \\ &= 3300 \text{ cal}\end{aligned}$$

Step II:

Calculate the value of  $\Delta G$  as follows:

$$\begin{aligned}\text{Then, } \Delta G &= \Delta H - T\Delta S \\ &= 3300 - 300 \times 20 \\ &= -2700 \text{ cal or } -2.7 \text{ kcal}\end{aligned}$$

Hence, **option 2** is correct.



**Q72)** Molar heat capacity or molar specific heat capacity is the amount of heat energy required to raise the temperature of 1 mole of a substance.

In SI units, molar heat capacity (symbol:  $c_m$ ) is the amount of heat in joules required to raise 1 mole of a substance 1 kelvin

$$c_m = Q/\Delta T$$

where  $Q$  is heat and  $\Delta T$  is the change in temperature. For most purposes, heat capacity is reported as an intrinsic property meaning it is a characteristic of a specific substance. Heat capacity is measured using a calorimeter.

Thermal capacity, also known as heat capacity, is a property of a material or system that measures the amount of heat required to change its temperature by a certain amount. Specifically, it is defined as the amount of heat energy needed to raise the temperature of a given quantity of the substance by one degree Celsius (or one Kelvin).

## Key Concepts:

### 1. Heat Capacity (C):

- **Definition:** The amount of heat required to change the temperature of an object or substance by one degree.
- .
- **Units:** Joules per degree Celsius ( $J/^{\circ}C$ ) or Joules per Kelvin ( $J/K$ ).

### 2. Specific Heat Capacity (c):

- **Definition:** The heat capacity per unit mass of a material. It measures how much heat is required to raise the temperature of one kilogram of the substance by one degree Celsius (or one Kelvin).
- 
- **Units:** Joules per kilogram per degree Celsius ( $J/kg^{\circ}C$ ) or Joules per kilogram per Kelvin ( $J/kg\cdot K$ ).

### 3. Molar Heat Capacity:

- **Definition:** The heat capacity per mole of a substance. It measures the amount of heat required to raise the temperature of one mole of the substance by one degree Celsius (or one Kelvin).
- 
- **Units:** Joules per mole per degree Celsius ( $J/mol^{\circ}C$ ) or Joules per mole per Kelvin ( $J/mol\cdot K$ ).

**Q73)** Explanation:

**Hint:** Work done is zero in the absence of external pressure.

**Explanation:**

(1) It is given that the ideal gas is present in a thermally insulated vessel. We know that in a thermally insulated vessel, there is no change in mass and temperature. So, this means that the number of moles will remain constant.

Further, there is no change in heat. Thus,  $q = 0$ .

(3) It is also given that  $P_{ext} = 0$ . So,  $w = -P_{ext}dV = 0$ . From the first law of thermodynamics we know that,  $\Delta U = q + w = 0$  This means the change in temperature is also not there because  $\Delta U = nC_v \Delta T$  So,  $T_2 = T_1$

(2) From Ideal gas equation, we know that  $PV = nRT$ . In the given conditions, the number of moles, i.e., 'n' is constant, the temperature is constant and R is already a gas constant. So, the right hand side is constant. There is a change in only pressure and volume in the above reaction.

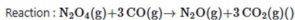
$$\text{Hence, } P_2V_2 = P_1V_1$$

(3) The equation  $P_2V_2^{\gamma} = P_1V_1^{\gamma}$  is for adiabatic processes. There is nothing mentioned about the adiabatic process in this question. Hence, it is an incorrect option.

Therefore, **option 1** is the correct answer.

**Q74) Explanation:**

Step 2: Calculate  $\Delta_r H$  for the given reaction



$$\begin{aligned}\Delta_r H &= [\Delta_f H (N_2O(g)) + 3 \Delta_f H (CO_2(g))] - [\Delta_f H (N_2O_4(g)) + 3 \Delta_f H (CO(g))] \\ &= \{ 81 + 3 (-393) - [9.7 + 3 (-110)] \} = -777.7 \text{ kJ mol}^{-1}\end{aligned}$$

Hence, the value of  $\Delta_r H$  for the reaction is  $-777.7 \text{ kJ mol}^{-1}$

**Q75) Explanation:**

Hint: Use the formula, that is,  $W = -p_{\text{ext}} (V_2 - V_1)$

In the first case, as the expression is against constant external pressure

$$\begin{aligned}W &= -p_{\text{ext}} (V_2 - V_1) = -2 \text{ bar} \times (50 - 10) \text{ L} \\ &= -80 \text{ L bar} \quad (1 \text{ L bar} = 100 \text{ J})\end{aligned}$$

$$W = -80 \times 100 = -8000 \text{ J}$$

$$1 \text{ kJ} = 1000 \text{ J}$$

$$W = -8 \text{ kJ}$$

**Q76) Explanation:**

**Step 1:** It is given that,  $\Delta_r H_{298 \text{ K}} = -54.07 \text{ kJ mol}^{-1}$ ,  $\Delta_r S_{298 \text{ K}}^\circ = 10 \text{ JK}^{-1} \text{ mol}^{-1}$  and  $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$ ,

$$\text{Temperature} = 298 \text{ K} \text{ and } 2.303 \times 8.314 \times 298 = 5705$$

This question aims to find  $\log_{10} K$

$$\text{We know that, } \Delta_r G^\circ = \Delta_r H^\circ - T \Delta_r S^\circ$$

$$\text{Hence, } \Delta G^\circ = -54.04 \times 10^3 \text{ J mol}^{-1} - (298 \text{ K} \times 10 \text{ JK}^{-1} \text{ mol}^{-1})$$

$$\Delta G^\circ = -57050 \text{ J mol}^{-1}$$

**Step 2:** It is also known that,  $\Delta G^\circ = -RT \ln K$  or  $\Delta G^\circ = -2.303 \times RT \log_{10} K$

$$\log_{10} K = -\frac{\Delta G^\circ}{2.303 \times RT}$$

$$\log_{10} K = -\frac{\Delta G^\circ}{2.303 \times 8.314 \times 298}$$

$$\log_{10} K = -\frac{-57050 \text{ J mol}^{-1}}{5705 \text{ J mol}^{-1}} = 10$$

Hence, **option 2** is the correct answer.

**Q77) Explanation:**

Hint: Follow Hess' Law

Explanation:



The bond energies of  $O-F$ ,  $O-H$ ,  $H-F$  and  $O=O$  as 44, 111, 135, and 119 kcal mol<sup>-1</sup> respectively

Step 1:  $\Delta H$  for the reaction is calculated by using

From Hess' Law

$$\Delta H = \sum B \cdot E_R - \sum B \cdot E_P$$

$$2(B.E_{O-H}) + 2(B.E_{O-F}) - 2(B.E_{O-F}) - B.E_{O=O}$$

On putting values we get,

$$(2 \times 44 + 2 \times 111) - (119 + 2 \times 135) = -79 \text{ kcal}$$

Hence, option 4 is the correct answer.

**Q78) Hint:  $w = -2.303nRT \log \frac{P_1}{P_2}$**

The formula of work done is as follows:

$$w = -2.303nRT \log \frac{P_1}{P_2}$$

The given values are as follows:

$$n = 1 \text{ mole}$$

$$P_1 = 5$$

$$P_2 = 1$$

$$R = 8.314$$

$$T = 300 \text{ K}$$

Calculate the value of work done as follows:

$$\begin{aligned} w &= -2.303 \times 1 \times 8.314 \times 300 \log \frac{5}{1} \\ &= -4014.6 \text{ J} \end{aligned}$$

Hence, the correct option is the fourth option.

**Q79) Explanation:**

According to Gibb's Helmholtz reaction for spontaneity,

$$\Delta G = \Delta H - T\Delta S$$

For the reaction to be spontaneous,  $\Delta G$  = negative.

For this,  $\Delta H$  should be negative and  $\Delta S$  should be positive.

$$\therefore \Delta H < 0 \text{ and } \Delta S > 0.$$

and also  $\Delta S = 0$  shows  $\Delta G$  a negative quantity.

**Q80) Explanation:**

In  $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ , entropy change is positive. In option d, A solid reactant gives a gaseous product. Hence the entropy change is expected to be maximum for this process.

**Q81) Explanation:**

Hint:  $\Delta G = -ve$  means spontaneous reaction.

For spontaneity means,  $\Delta G = -ve$

$$\Delta G = \Delta H - T\Delta S$$

$\Delta H$ , for endothermic process +ve

Entropy decreases as temperature decreases. At lower temperature,  $\Delta S$  is low hence, overall value of  $\Delta G = +ve$ .

But at high temperature T value and  $\Delta S$  value both increases.  $T\Delta S$  became more negative and will be greater than  $\Delta H$ , hence  $\Delta G = -ve$ , spontaneous

**Q82) Explanation:**

Hint:  $W = -P_{\text{ext}}\Delta V$

The gas expands it means that the initial volume is less than the final volume.

$$\begin{aligned} W &= -P_{\text{ext}}[V_f - V_i] \\ &= -10^5 [10^{-2} - 10^{-3}] \\ &= -10^5 \times 0.009 \\ &= -900 \text{ J} \end{aligned}$$

Hence, option 2 is the correct answer.

**Q83) Hint: Total entropy must be greater than zero**

The total entropy change ( $\Delta S_{\text{total}}$ ) for the system and surroundings of a spontaneous process is given by

$$\Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$$

**Q84) Explanation:**

Hint: In adiabatic conditions,  $q = 0$ .

And for free expansion;  $P_{\text{external}} = 0$   $W = -P_{\text{external}} \times V = 0 \times V = 0$   $w = 0$  From law of thermodynamics  $\Delta q + w = 0 + 0 = 0$   $\Delta U = 0$ ,  $\Delta U = nC_V \Delta T$

Q85) **Hint:**  $\Delta H = \Delta U + \Delta n_g RT$

**Explanation:**

**STEP 1:** The given reaction is:  $\text{NH}_2\text{CN (g)} + \frac{3}{2} \text{O}_2 \text{ (g)} \rightarrow \text{N}_2 \text{ (g)} + \text{CO}_2 \text{ (g)} + \text{H}_2\text{O (l)}$

$$\Delta U = -740 \text{ kJ mol}^{-1} \text{ and } T = 300\text{K}$$

$$\text{We know that, } \Delta H = \Delta U + \Delta n_g RT$$

where  $\Delta n_g$  = Total number of moles of gaseous products - Total number of moles of gaseous reactants.

$$\text{So, for the given reaction, } \Delta n_g = 2 - 2.5 = -0.5$$

$$\text{STEP 2: We know that, } \Delta H = \Delta U + \Delta n_g RT$$

$$\Delta H = -740 + \left( \frac{-0.5 \times 8.314 \times 300}{1000} \right) \text{ kJ mol}^{-1} = -741.2 \text{ kJ mol}^{-1}$$

Q86) **Explanation:**

$$\text{Hint: } \Delta G = \Delta H - T\Delta S$$

**Explanation:**

$$\text{STEP 1: Given, } \Delta H^\circ = -60.0 \text{ kJ and } \Delta S^\circ = -0.200 \text{ kJ K}^{-1}$$

$$\text{The value of free energy and entropy change: } \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\text{For a spontaneous reaction, the value of } \Delta G^\circ < 0$$

STEP 2: When the free energy change is 0, then the temperature can be calculated as:

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\text{Or, } 0 = -60.0 \text{ kJ} - T(-0.200 \text{ kJ mol}^{-1})$$

$$\text{Or, } \frac{60.0 \text{ kJ}}{0.200 \text{ kJ K}^{-1}} = T$$

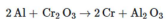
$$\text{Or, } T = 300 \text{ K}$$

Thus, the negative free energy change will be negative when the temperature (T) is less than 300 K.

Hence, **option 1** is the correct answer.

Q87) **Explanation:**

$$\text{Hint: } \Delta H_r^\circ = [2 \Delta H_f^\circ (\text{Cr}) + \Delta H_f^\circ (\text{Al}_2\text{O}_3)] - [2 \Delta H_f^\circ (\text{Al}) + \Delta H_f^\circ (\text{Cr}_2\text{O}_3)]$$



$$\Delta H_r^\circ = [2 \Delta H_f^\circ (\text{Cr}) + \Delta H_f^\circ (\text{Al}_2\text{O}_3)] - [2 \Delta H_f^\circ (\text{Al}) + \Delta H_f^\circ (\text{Cr}_2\text{O}_3)]$$

$$\Delta H_r^\circ = [0] + (-1596 \text{ kJ}) - [0] + (-1134)$$

$$\Delta H_r^\circ = -1596 \text{ kJ} + 1134 \text{ kJ}$$

$$\Delta H_r^\circ = -462 \text{ kJ}$$

**Q88) Explanation:**

**Hint:**  $\Delta U = nC_V\Delta T$

**Explanation:**

Step 1: Number of moles of ideal gas = 5

Temperature = 100 K

Also,  $C_V = 28 \text{ J K}^{-1}$

So,  $\Delta U = nC_V \Delta T = 5 \text{ moles} \times 28 \text{ J K}^{-1} \times 100 \text{ K}$

$\Delta U = 14000 \text{ J} = 14 \text{ kJ}$

Step 2: If  $T_1 = 100 \text{ K}$ ,  $T_2 = 200 \text{ K}$  and  $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

So,  $\Delta(pV) = nR(\Delta T)$

or,  $\Delta(pV) = 5 \times 8.314 \text{ J mol}^{-1} \text{ K}^{-1}(200 \text{ K} - 100 \text{ K})$

or,  $\Delta(pV) = 4157 \text{ J} = 4.157 \text{ kJ}$

Hence, **option 3** is the correct answer.

**Q89) Explanation:**

**Hint:** In the combustion process, heat is released.

**Explanation:**

(i) When the temperature increases, the randomness of the molecules increases, due to which the entropy of the system increases. So, the entropy change is possible at higher temperatures.

(ii) Combustion reactions are almost always exothermic in nature. Thus, the combustion of methane is an exothermic process.

(iii) The First Law of Thermodynamics states that energy can be converted from one form to another by interacting with heat, work, and internal energy. Still, it cannot be created or destroyed, under any circumstances.

Mathematically, this is represented as:  $q = \Delta U + w$ . For a cyclic process, the change in internal energy is zero,  $q = w$ .

(iv) A process without the transfer of heat to or from a system, so that  $q = 0$ , is called adiabatic.

Only statement IV is the correct statement. Hence, **option 4** is the correct answer.

Q90) Hint: Extensive properties depends on the quantity and intensive property does not

Step 1:

**Extensive properties** Those properties whose value depends on the quantity or size of matter present in the system is known as extensive properties.

e.g., mass, internal energy, heat capacity.

**Intensive properties** Those properties which do not depend on the quantity or size of matter present are known as intensive properties. e.g., pressure, molar heat capacity, density, mole fraction, specific heat, temperature and molarity.

Step 2:

Mole fraction or molarity of a solution is same whether we take a small amount of the solution or large amount of the solution.

Ratio of two extensive properties is always intensive.

$$\frac{\text{Extensive}}{\text{Extensive}} = \text{Intensive}$$

So, mole fraction and molarity are intensive properties.

$$\text{e.g., mole fraction} = \frac{\text{Moles of the component}}{\text{Total no. of mole}} = \frac{\text{Extensive}}{\text{Extensive}}$$

$$\text{and molarity} = \frac{\text{Mole}}{\text{Volume}} = \frac{\text{Extensive}}{\text{Extensive}}$$



# **Mitoprep**

## **Sapno Ki Udaan**